

CHECKPOINT REPORT

Predictive Maintenance for Metro Climate & Air Systems

Data understanding and pre-processing complete. Modelling underway.

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MetroPT-3 Air Production Unit · 1,516,948 readings

4 of 8

weeks elapsed (to 18 Jul)

16,773

modelling windows

0.58

best F1 so far

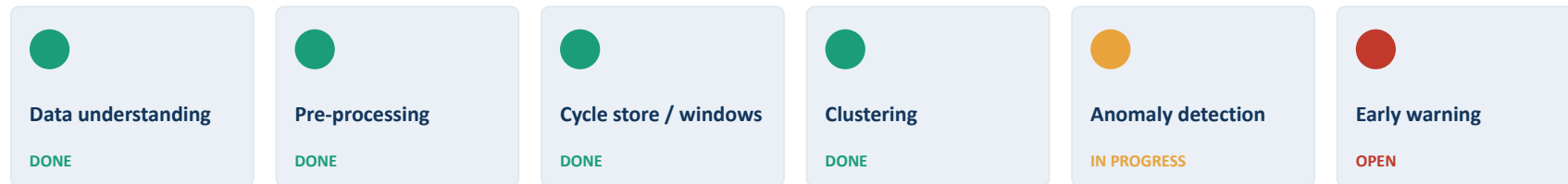
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events detected

Where the Pipeline Stands

PROGRESS

Four stages complete. The data reduction is the headline.



1.5M raw readings reduce to 16,773 analysable windows: a 90x reduction that preserves the cycle and duty statistics a leak acts on

Three Cleaning Decisions, Three Reasons

CLEANING

Each choice was made against a specific property of this sensor stream.

1

Flag gaps, never interpolate

909 hours of the 213-day record are missing across 331 gaps. Interpolating would invent smooth, well-behaved data exactly where the machine was out of service, teaching the model that the smoothest data is the most normal.

2

Correct offsets only where zero is expected

TP2 and DV_pressure rest at -0.012 and -0.020 bar: transducer bias. H1 rests at 8.64 bar because that is a real pressure. A blanket idle-median correction would have destroyed H1 entirely.

3

Median filter, not mean

A 5-sample median removes transient spikes while preserving the sharp edges of state transitions. A mean filter smears those edges, and the edges are what define a compression cycle.

909 h

of record missing, now flagged

0.11%

sustained frozen sensor holds

97k/s

cleaning throughput

The Compression Cycle Broke as a Unit of Analysis

FINDING 1

The proposal was wrong about this, and the reason is physically interesting.

What we expected

Aggregate 1.5M readings into ~16,900 compression cycles and model each cycle. A leak shows up as longer refills and shorter intervals.

10,974

cycles found in the cleaned record

What actually happened

During a failure the compressor stops cycling and runs continuously. There are almost no cycle boundaries inside the events we most need to model.

3

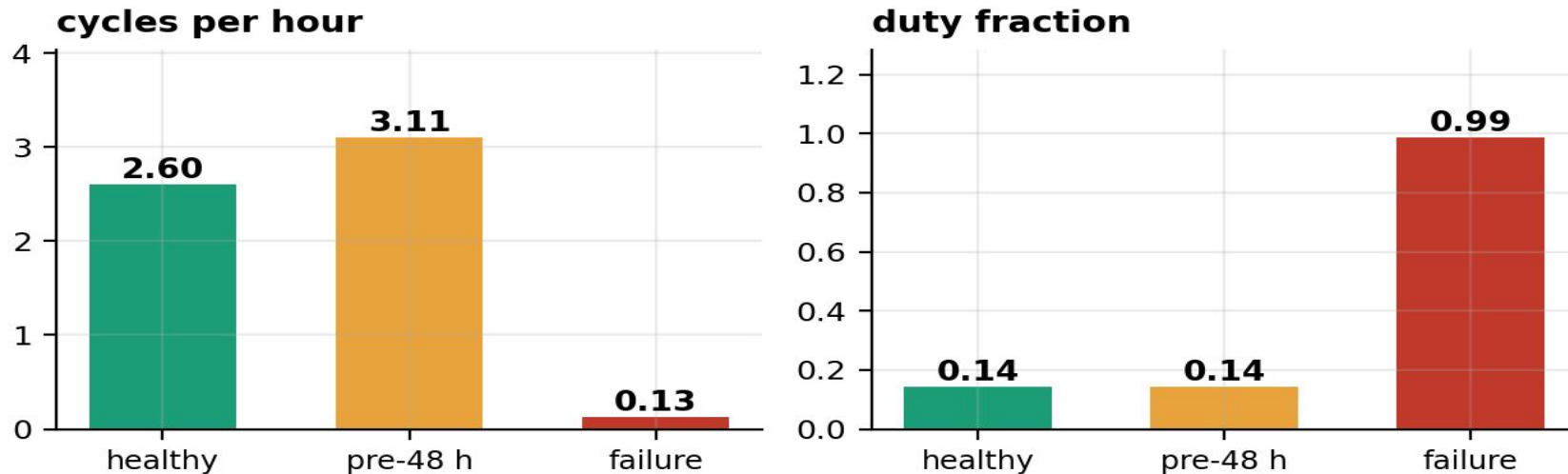
of those cycles begin inside a failure window

The fix, and why it is better: the modelling unit is now a fixed 15-minute window, with cycle statistics carried in as features. Cycles still describe the physics, but the window never disappears when the machine changes behaviour. The absence of cycling is now itself a measurable feature rather than a hole in the data.

A Three-Stage Degradation Signature

FINDING 2

Cycling frequency moves first. Duty cycle moves only at onset.



HEALTHY

2.60 cycles/h



PRE-FAILURE (48 h)

3.11 cycles/h (+20%)



FAILURE

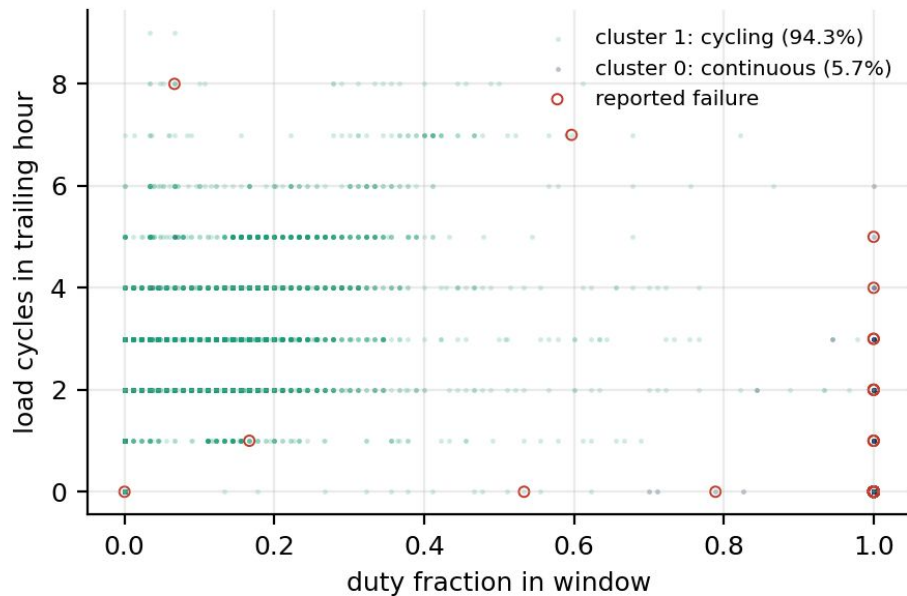
0.13 cycles/h (-95%)

Why this matters: duty cycle is flat at 0.14 right up to onset, so any duty-based alarm has zero lead time. The early signal lives in cycle frequency.

Unsupervised Structure Confirms Two Regimes

CLUSTERING

$k = 2$ chosen by silhouette (0.676). No label was used to find this.



Cluster 1: cycling

94.3% of windows · duty 0.11, 2.6 cycles/h

0.03% failure rate

Cluster 0: continuous run

5.7% of windows · duty 1.00, 0.004 cycles/h

34.0% failure rate

98.5% of failure windows land in cluster 0

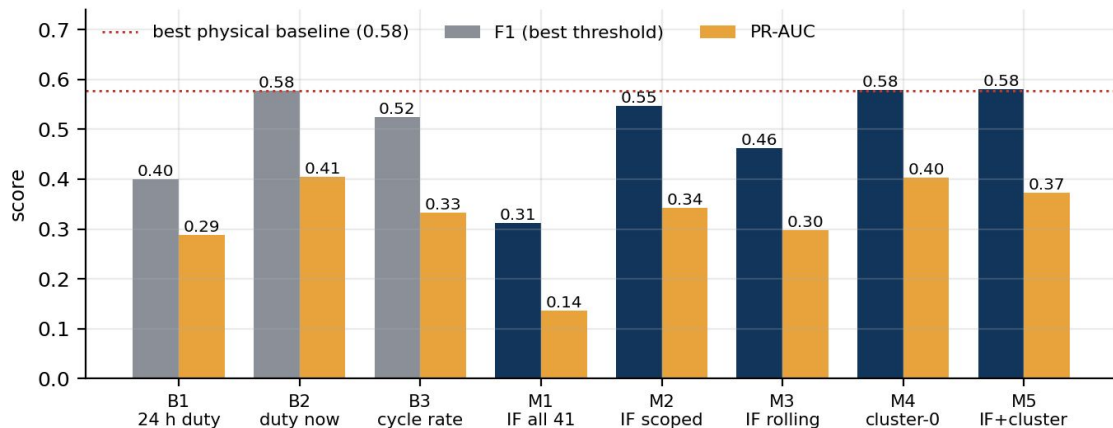
But only 34% of cluster 0 is a reported failure. Continuous running is necessary, not sufficient.

The clustering found the fault regime without being told it existed. That is the strongest validation we have that the feature set encodes the physics.

The Sophisticated Model Lost to the Simple One

RESULTS

We could have retuned until the numbers improved. Diagnosing the failure was worth more.



Diagnosis

Isolation Forest on all 41 features scored worse than a single duty threshold (F1 0.312 vs 0.577)

Cause: mean duty rises 50% from the Feb-Mar training window to the test period. The detector spends its alarm budget on seasonal drift, not on faults.

0.312 → 0.546

F1 after scoping to 14 physics-based features

0.58

three methods tie here / gated model preferred for its tunable score

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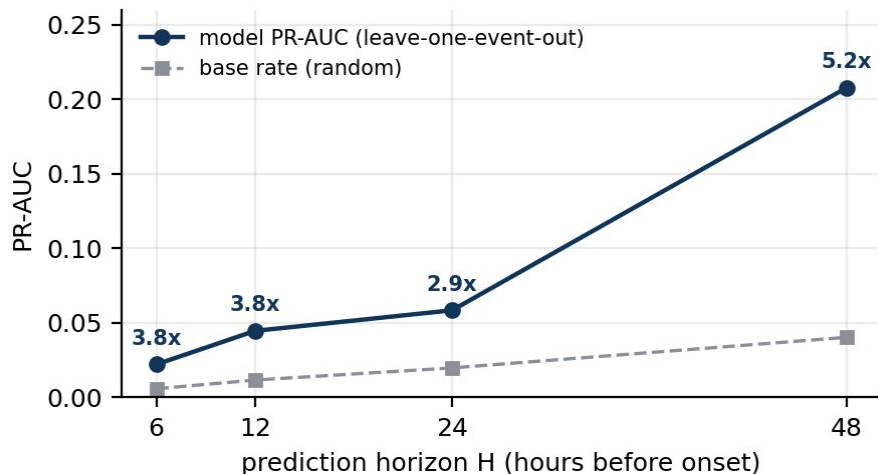
events detected by every model that we tested

Conclusion: concurrent detection is effectively solved by physics. Adding model complexity buys almost nothing. So the remaining research value is not here.

The Real Remaining Problem: Lead Time

OPEN QUESTION

Detection at onset is easy. Prediction before onset is not.



How this was tested

Every window inside a failure was removed, so the model cannot cheat by detecting the event it is meant to predict.

Validation is leave-one-event-out, because four events cannot support a random split.

PR-AUC is reported against base rate, since at 0.6% prevalence a raw PR-AUC of 0.02 can still be a 3.8x lift.

Where this leaves us: the model beats chance at every horizon (2.9x to 5.2x lift) and flags all four events, but absolute precision is low. There is real early signal and it is not yet strong enough to act on. Closing that gap is the work of the remaining weeks.

Challenges and the Backup Plan

RISKS

Named now, with a decision rule for when to switch.

1

Label incompleteness

Two thirds of cluster 0 is unreported. Precision is capped by the labels, not the model.

Mitigation: Report unlabelled high-score episodes separately, against confirmed events.

2

Four events only

Every supervised number rests on four positives. Variance dominates point estimates.

Mitigation: Leave-one-event-out throughout. Report spread, never a bare mean.

3

Concept drift

Mean duty rises 50% from training to test period and has already broken one model.

Mitigation: Scoping to 14 physical features, which recovered F1 from 0.312 to 0.546. A rolling 30-day reference was also tried and underperformed; see report §9.1.

4

Early warning may not reach usable precision

Current lift is real but precision is low at every horizon.

Mitigation: Fall back to change-point detection on the CYCLE-RATE series, not duty: duty carries no pre-onset signal, so a duty-based change point would inherit zero lead time.

Switch rule: if week 6 leave-one-event-out PR-AUC lift stays below 3x at the 24-hour horizon, we stop tuning and ship the change-point fallback.

Four Weeks to the Finish Line

Scope deliberately narrowed to the one question that is still open.

19 July to 15 August 2027

WEEK 5 · 19-25 JUL

Early-warning features

Build cycle-frequency and pressure-recovery features targeted at the pre-failure stage.

WEEK 6 · 26 JUL-1 AUG

Model and decide

Gradient boosting at 24h and 48h horizons. Apply the switch rule at week's end.

WEEK 7 · 2-8 AUG

Efficiency and ablations

Full-stream throughput, per-window latency, feature ablations.

WEEK 8 · 9-15 AUG

Write up and ship

Final report, reproducible repository, dashboard mock.

Dropped from scope: the LSTM autoencoder. Isolation Forest already showed that added capacity does not beat physics on this signal, so the deep model would consume a week to answer a question we have answered.

The big picture: every task above serves one outcome, giving a maintenance planner enough warning to pull a train before it strands passengers mid-line.